

Congestion Control & Quality of services (QoS)

Content

Congestion Control Techniques
Congestion Prevention Policies
Quality of services (QoS)

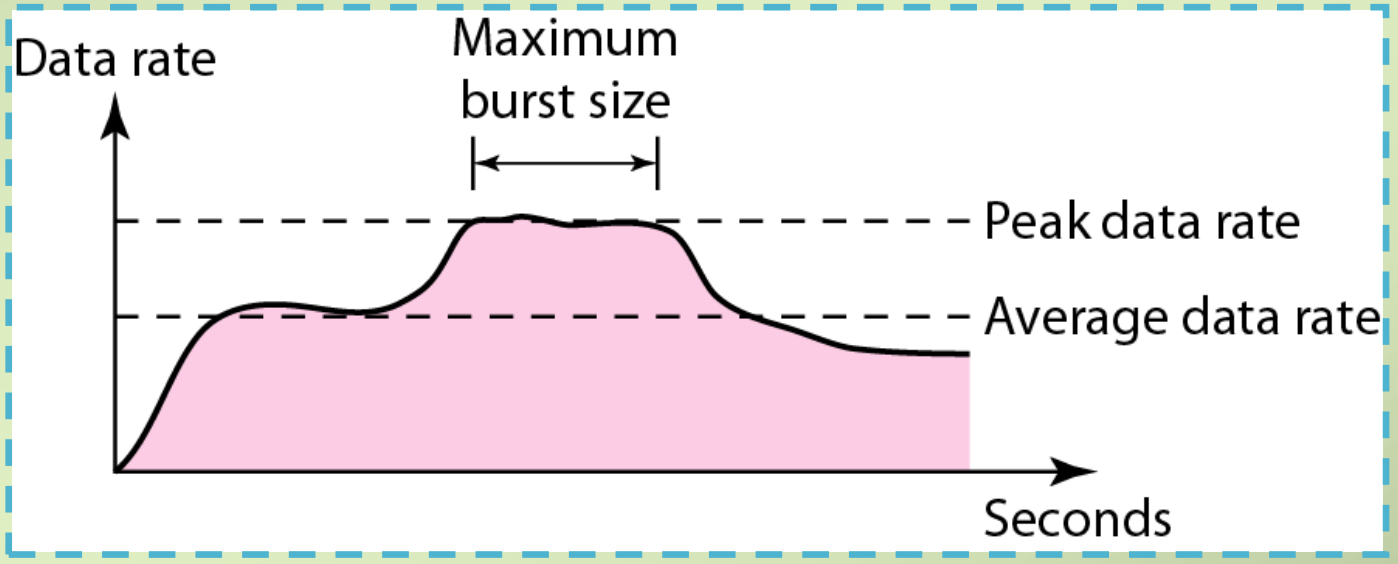


Introduction

- *The main focus of congestion control and quality of service is data traffic.*
- *In congestion control - to avoid traffic congestion.*
- *In quality of service, an appropriate environment for the traffic is created*

Traffic descriptors

Traffic descriptors are qualitative values that represent a data flow.



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Average Data Rate:

- The average data rate is the number of bits sent during a period of time, divided by the number of seconds in that period.
- The average data rate is a very useful characteristic of traffic which indicates the average bandwidth needed by the traffic.

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Peak Data Rate

- The peak data rate defines the maximum data rate of the traffic.
- The peak data rate is a very important measurement because it indicates the peak bandwidth that the network needs for traffic to pass through without changing its data flow.

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Maximum Burst Size

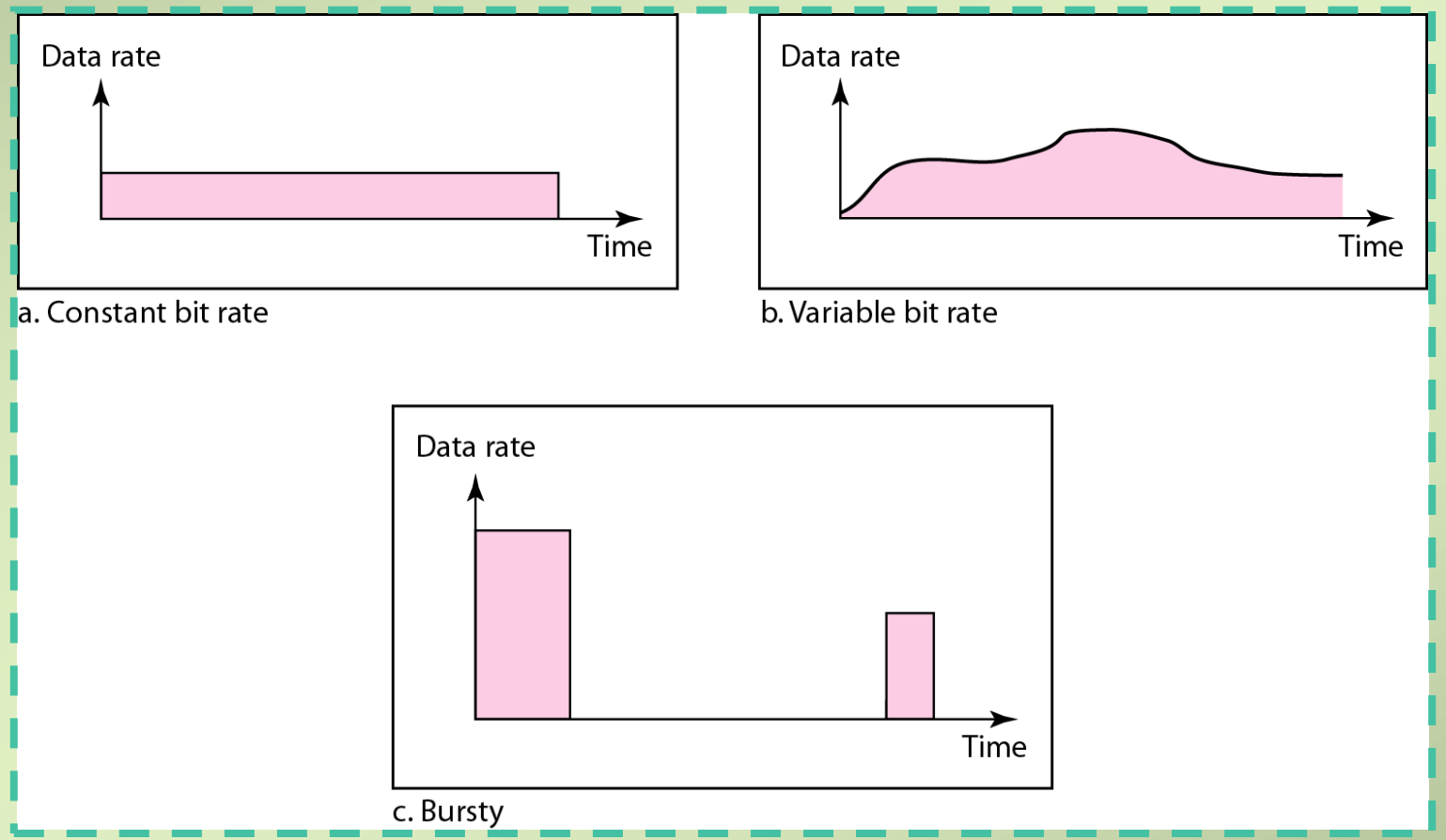
- The peak data rate is a critical value for the network, it can usually be ignored if the duration of the peak value is very short.
- For example, if data are flowing steadily at the rate of 1 Mbps with a sudden peak data rate of 2 Mbps for just 1ms, the network probably can handle the situation.
- The maximum burst size normally refers to the maximum length of time the traffic is generated at the peak rate.

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Effective Bandwidth

- The effective bandwidth is the bandwidth that the network needs to allocate for the flow of traffic.
- The effective bandwidth is a function of three values:
  - Average data rate
  - Peak data rate
  - Maximum burst size
- The calculation of this value is very complex.

Three traffic profiles



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Constant Bit Rate

- A constant-bit-rate (CBR), or a fixed-rate, traffic model has a data rate that does not change. In this type of flow, the average data rate and the peak data rate are the same.
- The maximum burst size is not applicable. This type of traffic is very easy for a network to handle since it is predictable.
- The network knows in advance how much bandwidth to allocate for this type of flow.

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Variable Bit Rate:

- In the variable-bit-rate (VBR) category, the rate of the data flow changes in time, with the changes smooth instead of sudden and sharp.
- In this type of flow, the average data rate and the peak data rate are different. The maximum burst size is usually a small value.
- This type of traffic is more difficult to handle than constant-bit-rate traffic, but it normally does not need to be reshaped.

Contd...

Bursty

- In the bursty data category, the data rate changes suddenly in a very short time. It may jump from zero, for example, to 1 Mbps in a few microseconds and vice versa.
- It may also remain at this value for a while. The average bit rate and the peak bit rate are very different values in this type of flow.
- The maximum burst size is significant. This is the most difficult type of traffic for a network to handle because the profile is very unpredictable.
- Bursty traffic is one of the main causes of congestion in a network

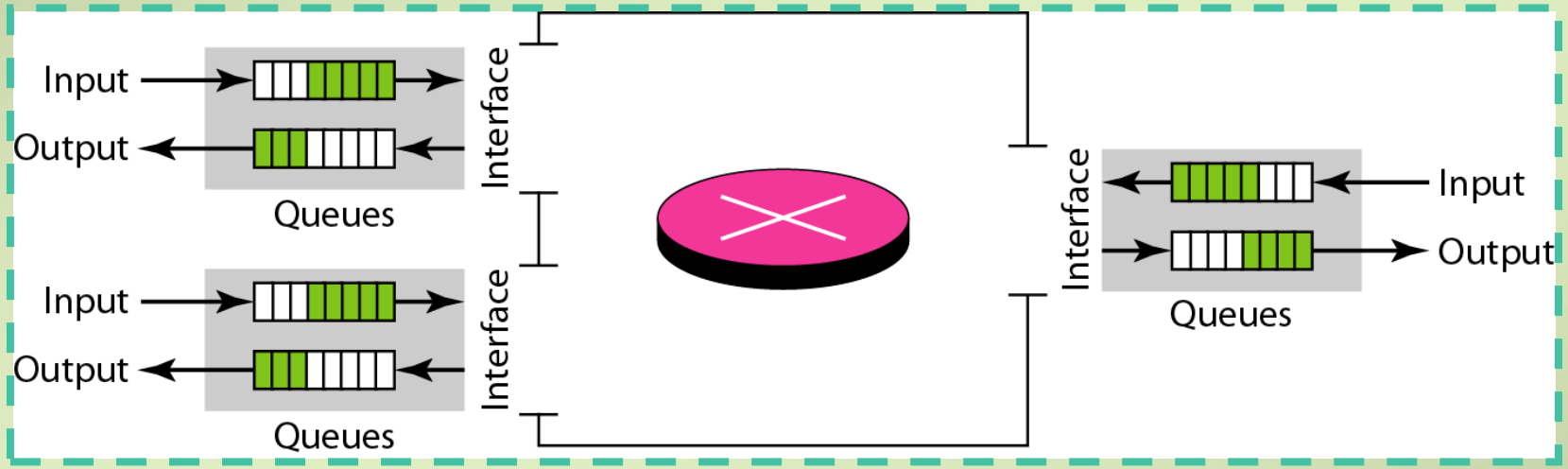
Congestion

- Congestion in a network may occur if the load on the network—the number of packets sent to the network—is greater than the capacity of the network—the number of packets a network can handle.
- Congestion control refers to the mechanisms and techniques to control the congestion and keep the load below the capacity.
- Congestion in a network or internetwork occurs because routers and switches have queues-buffers that hold the packets before and after processing.

Cond…

- A router, for example, has an input queue and an output queue for each interface.
- When a packet arrives at the incoming interface, it undergoes three steps before departing
  - The packet is put at the end of the input queue while waiting to be checked.
  - The processing module of the router removes the packet from the input queue once it reaches the front of the queue and uses its routing table and the destination address to find the route.
  - The packet is put in the appropriate output queue and waits its turn to be sent

Queues in a router



Network Performance

- Congestion control involves two factors that measure the performance of a network: delay and throughput

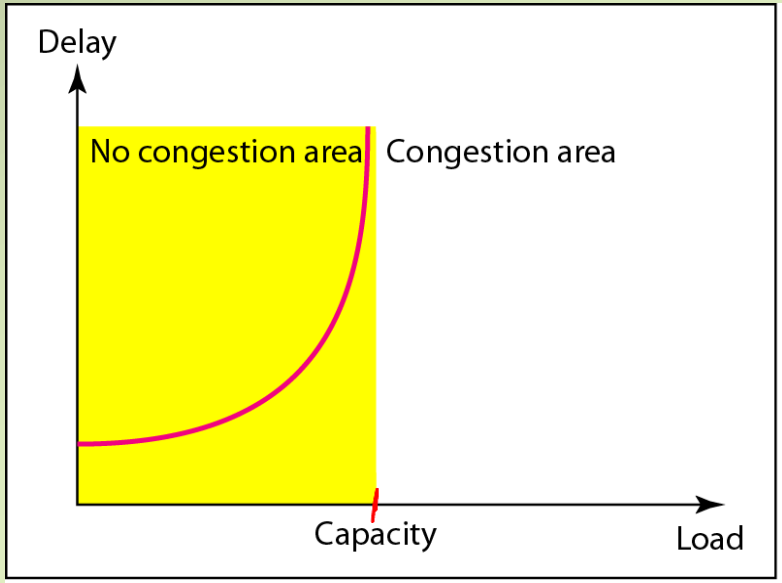
Delay Vs Load

- when the load is much less than the capacity of the network, the delay is at a minimum.
- This minimum delay is composed of propagation delay and processing delay, both of which are negligible.
- when the load reaches the network capacity, the delay increases sharply because we now need to add the waiting time in the queues (for all routers in the path) to the total delay.

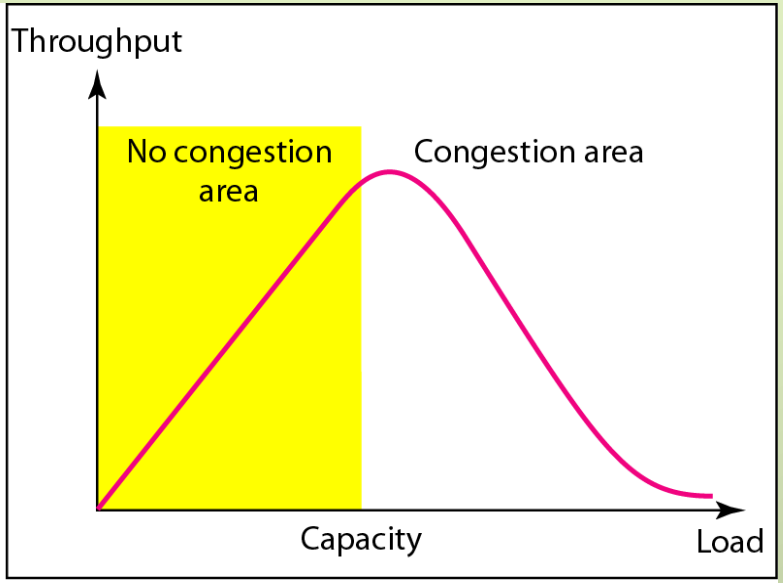
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- The delay becomes infinite when the load is greater than the capacity.
- If this is not obvious, consider the size of the queues when almost no packet reaches the destination, or reaches the destination with infinite delay; the queues become longer and longer.
- Delay has a negative effect on the load and consequently the congestion.
- When a packet is delayed, the source, not receiving the acknowledgment, retransmits the packet, which makes the delay, and the congestion, worse

Packet delay and throughput as functions of load



a. Delay as a function of load



b. Throughput as a function of load

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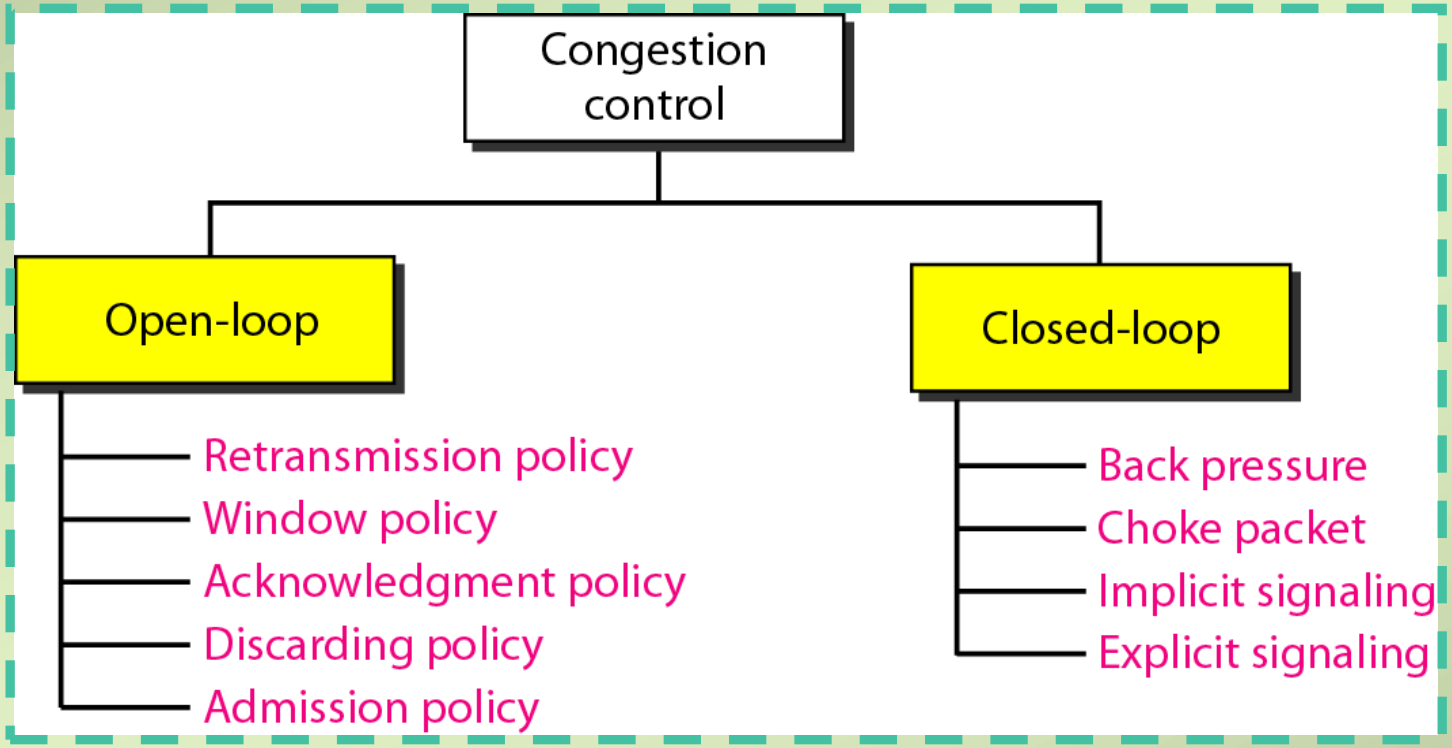
Throughput Versus Load

- when the load is below the capacity of the network, the throughput increases proportionally with the load
- We expect the throughput to remain constant after the load reaches the capacity, but instead the throughput declines sharply.
- The reason is the discarding of packets by the routers. When the load exceeds the capacity, the queues become full and the routers have to discard some packets.
- Discarding packet; does not reduce the number of packets in the network because the sources retransmit the packets, using time-out mechanisms, when the packets do not reach the destinations

CONGESTION CONTROL

- Congestion control refers to techniques and mechanisms that can either prevent congestion, before it happens, or remove congestion, after it has happened.
- Two broad categories:
  - Open-loop congestion control (prevention)
  - Closed-loop congestion control (removal)

Congestion control categories



Open-Loop Congestion Control

- In open-loop congestion control, policies are applied to prevent congestion before it happens. In these mechanisms, congestion control is handled by either the source or the destination.

Retransmission Policy

- Retransmission is sometimes unavoidable. If the sender feels that a sent packet is lost or corrupted, the packet needs to be retransmitted.
- Retransmission increase congestion in the network. A good retransmission policy can prevent congestion.
- The retransmission policy and the retransmission timers must be designed to optimize efficiency and at the same time prevent congestion.

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Window Policy

- The type of window at the sender may also affect congestion. The Selective Repeat window is better than the Go-Back-N window for congestion control.
- In the Go-Back-N window, when the timer for a packet times out, several packets may be resent, although some may have arrived safe and sound at the receiver.
- This duplication may make the congestion worse. The Selective Repeat window, on the other hand, tries to send the specific packets that have been lost or corrupted.

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Acknowledgment Policy

- The acknowledgment policy imposed by the receiver may also affect congestion.
- If the receiver does not acknowledge every packet it receives, it may slow down the sender and help prevent congestion.
- A receiver may send an acknowledgment only if it has a packet to be sent or a special timer expires.
- A receiver may decide to acknowledge only N packets at a time.
- Sending fewer acknowledgments means imposing less load on the network.

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Discarding Policy

- A good discarding policy by the routers may prevent congestion and at the same time may not harm the integrity of the transmission.
- For example, in audio transmission, if the policy is to discard less sensitive packets when congestion is likely to happen, the quality of sound is still preserved and congestion is prevented or alleviated.

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Admission Policy

- An admission policy, which is a quality-of-service mechanism, can also prevent congestion in virtual-circuit networks.
- Switches in a flow first check the resource requirement of a flow before admitting it to the network.
- A router can deny establishing a virtual circuit connection if there is congestion in the network or if there is a possibility of future congestion.

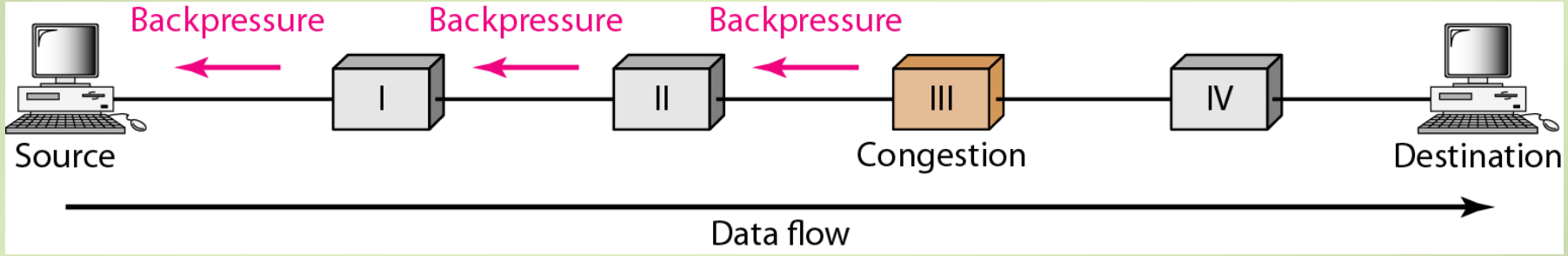
Closed-Loop Congestion Control

- Closed-loop congestion control mechanisms try to alleviate congestion after it happens.
- Several mechanisms have been used by different protocol.

Backpressure

- The technique of backpressure refers to a congestion control mechanism in which a congested node stops receiving data from the immediate upstream node or nodes.
- This may cause the upstream node or nodes to become congested, and they, in turn, reject data from their upstream nodes or nodes. And so on.
- Backpressure is a node-to-node congestion control that starts with a node and propagates, in the opposite direction of data flow, to the source.
- The backpressure technique can be applied only to virtual circuit networks, in which each node knows the upstream node from which a flow of data is coming.

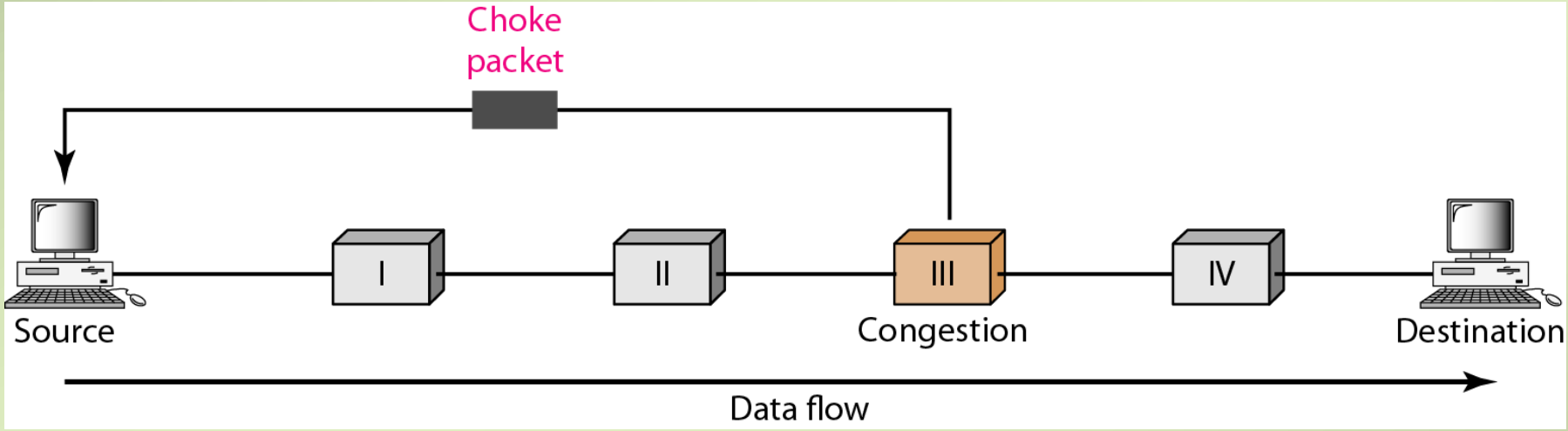
Backpressure method for alleviating congestion



Choke Packet

- A choke packet is a packet sent by a node to the source to inform it of congestion.
- In backpressure, the warning is from one node to its upstream node, although the warning may eventually reach the source station.
- In the choke packet method, the warning is from the router, which has encountered congestion, to the source station directly.
- The intermediate nodes through which the packet has traveled are not warned.
- The warning message goes directly to the source station; the intermediate routers, and does not take any action.

Choke packet



Implicit Signaling

- In implicit signaling, there is no communication between the congested node or nodes and the source.
- The source guesses that there is a congestion somewhere in the network from other symptoms.
- For example, when a source sends several packets and there is no acknowledgment for a while, one assumption is that the network is congested.
- The delay in receiving an acknowledgment is interpreted as congestion in the network; the source should slow down.

Explicit Signaling

- The node that experiences congestion can explicitly send a signal to the source or destination.
- The explicit signaling method, however, is different from the choke packet method.
- In the choke packet method, a separate packet is used for this purpose; in the explicit signaling method, the signal is included in the packets that carry data.

Backward Signaling

- A bit can be set in a packet moving in the direction opposite to the congestion.
- This bit can warn the source that there is congestion and that it needs to slow down to avoid the discarding of packets

Forward Signaling

- A bit can be set in a packet moving in the direction of the congestion. This bit can warn the destination that there is congestion.
- The receiver in this case can use policies, such as slowing down the acknowledgments, to lighten the congestion.

Congestion Prevention Policies

Layer	Policies
Transport	• Retransmission policy • Out-of-order caching policy • Acknowledgement policy • Flow control policy • Timeout determination
Network	• Virtual circuits versus datagram inside the subnet • Packet queueing and service policy • Packet discard policy • Routing algorithm • Packet lifetime management
Data link	• Retransmission policy • Out-of-order caching policy • Acknowledgement policy • Flow control policy

Practical – Lab Exercise

- 8. Implementation of IOT using cisco packet tracer
- 9. Basic IPV6 Addressing with cisco packet tracer

  (https://www.youtube.com/watch?v=Y3z-eK1jczc)
- 10. Implementation of ARP using cisco packet tracer

  (https://www.youtube.com/watch?v=M_vCHucoHlw)